



Department of Electrical and Computer Engineering

ECE 140: LINEAR CIRCUITS

Final Examination – Winter 2019

April 16, 2019

12:30-3:00 PM

Instructor: Mohamed-Yahia Dabbagh

Time Allowed: 2 hour 30 minutes

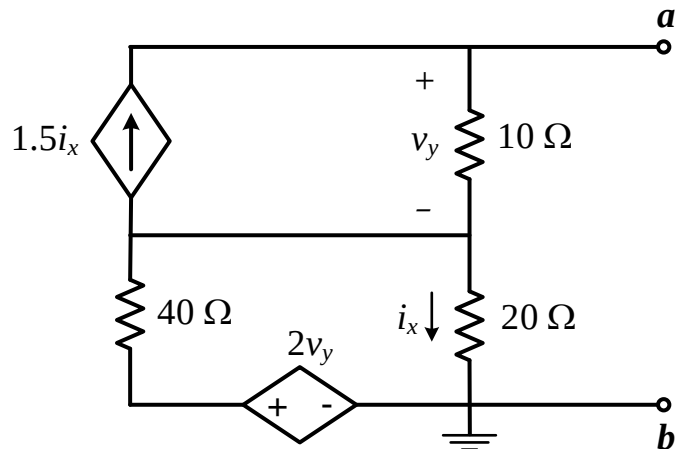
Student First Name:	Last Name:
Student ID Number:	
Lecture Section (check one): ☐ LEC 001 – ECE ☐ LEC 002 – SE	

Instructions:

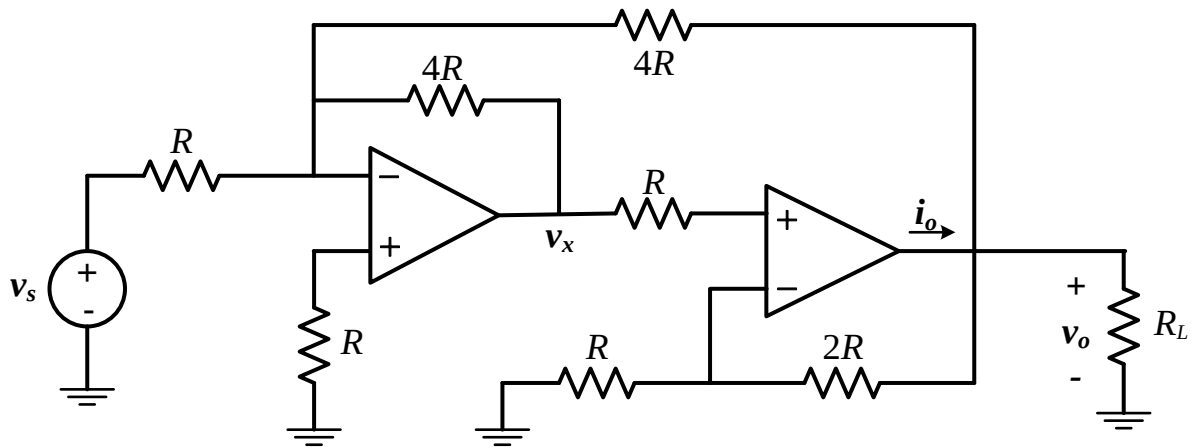
1. This exam has 22 pages, including this cover sheet. Please, keep all sheets stapled. You may detach the formula sheet only, which will not be marked. Please, write your name on it.
2. Answer all questions.
3. This is a closed book and notes examination.
4. Calculators with cleared memory are allowed. However, it is not allowed to solve a set of equations using a calculator. Show your work.
5. Place your WATCARD on the table; and sign the attendance sheet when provided by the proctor.
6. Clearly show all steps used in the solution process. No marks will be given for numerical results unless accompanied by a correct solution method.
7. If you need extra space, you can use the back of pages. Please, indicate this on the front.
8. Use correct units in all your answers.
9. No questions will be answered during the exam.
10. No student can leave the exam room in the first 75 minutes and the last 5 minutes.

Question	1	2	3	4	5	6	7	TOTAL
Mark	/12	/12	/15	/15	/17	/12	/7	/90
Marker								

Question 1: [12 Marks] Consider the circuit shown below. Find Thevenin equivalent circuit (v_{th} and R_{th}) between the terminals a - b .

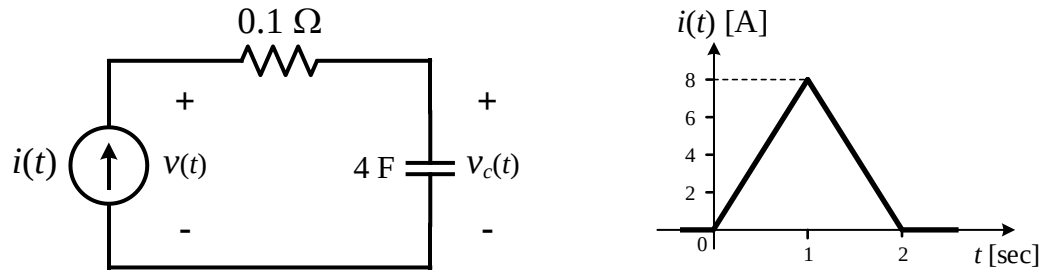


Question 2: [12 Marks] Assume *ideal op-amps* in the circuit shown below.



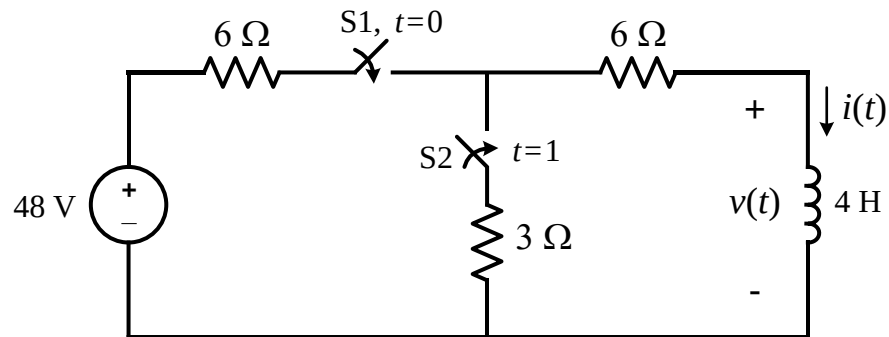
- (a) Find the voltage gain $G_v = \frac{v_o}{v_s}$, independent of R . Hint: Find v_o in terms of v_x first.
- (b) Find the value of R for which $v_o = -15$ V, $i_o = -20$ mA, and $R_L = 1$ k Ω .

Question 3: [15 Marks] Consider the following circuit and the current source function. Assume the capacitor is discharged at $t = 0$ s.



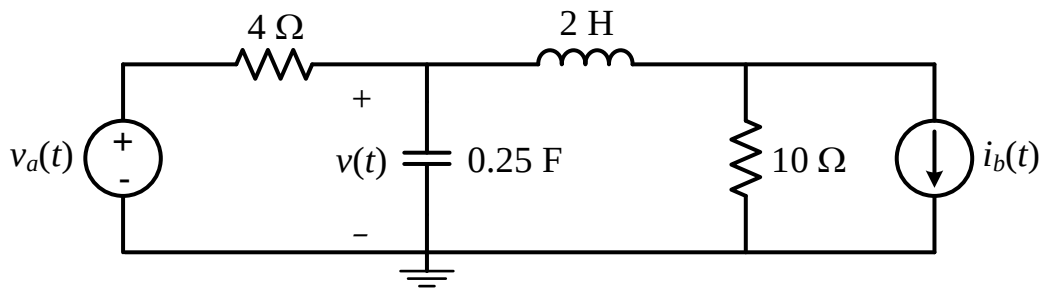
- (a) Find the voltage functions for $v_c(t)$ and $v(t)$ for $0 \leq t \leq \infty$.
- (b) Find the energy supplied by the current source from $t = 0$ to $t = \infty$.

Question 4: [15 Marks] For the circuit shown below, the switches S1 and S2 have been open before $t = 0$. The switch S1 is closed at $t = 0$ second, and S2 is closed at $t = 1$ second.



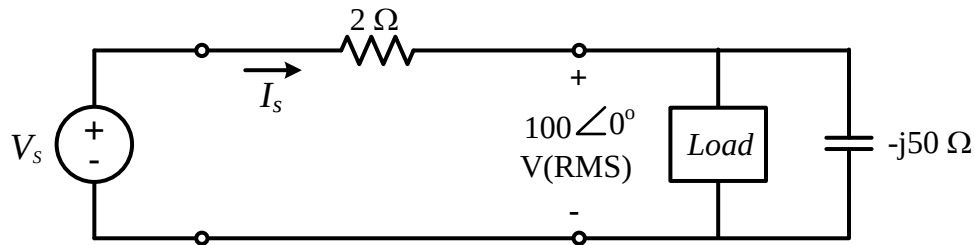
- (a) Find the inductor current $i(t)$ and the voltage $v(t)$ for $0 \leq t < 1$ second.
- (b) Find the inductor current $i(t)$ and the voltage $v(t)$ for $t \geq 1$ second.
- (c) Sketch $i(t)$ and $v(t)$ for all t . Identify the values at $t = 0$, $t = 1$ second, and $t \rightarrow \infty$.

Question 5: [17 Marks] The following circuit is in steady-state and has both DC and AC sources where $v_a(t) = 7 \text{ V}$ and $i_b(t) = 2 \cos(2t) \text{ A}$.



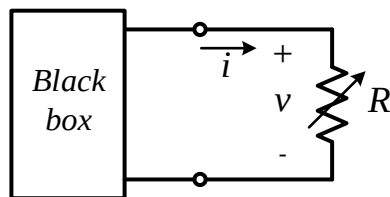
- (a) Find the voltage $v(t)$.
- (b) Find the average power absorbed by the 4Ω resistor.

Question 6: [12 Marks] A load and a capacitor are connected to a voltage source V_s through a cable that has a $2\ \Omega$ resistance, as shown below. The load absorbs 400 W at power factor $pf = 0.8$ lagging and voltage 100 V(RMS).



- (a) Find the cable current I_s and the source voltage V_s . Give your answers in polar form.
- (b) Find the supplied active power, reactive power, and apparent power for the source V_s .

Question 7: [7 Marks] A black box with a resistive circuit is connected to a variable resistor, as shown below. The voltage and current are measured for two values of the resistor, as shown in the table.



$R (\Omega)$	$v (\text{V})$	$i (\text{A})$
2	6	3
8	16	2

- Find Thevenin equivalent circuit for the black box.
- Find the maximum power that can be supplied by this black box.
- Find the value of the resistor R if $v = 24 \text{ V}$.

Name:**FORMULA SHEET** (This page will not be marked. You may detach it, but write your name)

- Inductor: $v(t) = L \frac{di}{dt}$

$$i(t) = \frac{1}{L} \int_{t_o}^t v(t) dt + i(t_o)$$

$$W = \frac{1}{2} L i^2$$

- Capacitor: $i(t) = C \frac{dv(t)}{dt}$

$$v(t) = \frac{1}{C} \int_{t_o}^t i(t) dt + v(t_o)$$

$$W = \frac{1}{2} C v^2$$

- Transient Analysis:

RC Circuit:

$$v(t) = v(\infty) + [v(t_o) - v(\infty)] e^{-\frac{t-t_o}{\tau}}$$

Where $\tau = RC$ RL Circuit:

$$i(t) = i(\infty) + [i(t_o) - i(\infty)] e^{-\frac{t-t_o}{\tau}}$$

Where $\tau = L/R$

- Sinusoids:

$$\sin(\theta) = \cos(\theta - 90^\circ)$$

$$-\sin(\theta) = \cos(\theta + 90^\circ)$$

$$-\cos(\theta) = \cos(\theta \pm 180^\circ)$$

- Impedance:

$$\mathbf{Z} = |\mathbf{Z}| \angle \xi = R + jX$$

$$\mathbf{Z}_L = j\omega L, \quad \mathbf{Z}_C = \frac{1}{j\omega C}$$

- Power:

$$\mathbf{S} = \frac{1}{2} \mathbf{V} \mathbf{I}^* = \mathbf{V}_{rms} \mathbf{I}_{rms}^* = |\mathbf{S}| \angle \xi = P + jQ$$

$$\begin{aligned} P &= V_{rms} I_{rms} \cos(\theta - \phi) \\ &= I_{rms}^2 R \end{aligned}$$

$$\begin{aligned} Q &= V_{rms} I_{rms} \sin(\theta - \phi) \\ &= I_{rms}^2 X \end{aligned}$$

$$pf = \frac{P}{|\mathbf{S}|} = \cos(\theta - \phi)$$

- The solution of $ax^2 + bx + c = 0$ is:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$